

PAPER_360 — J1610+1811 High-z Quasar Jet at z=6.5: Relativistic Lorentz Factor k_{rel} Coupling

Whitepaper Series: Star-Magic UQFF Phase 2

Session: 97

Source: gok_share_31b5c807a4.txt (Supplemental Gap Analysis Block)

Classification: FIRST UQFF high-z quasar jet (z=6.5) with Lorentz factor $k_{\text{rel}} = \Gamma^2$ relativistic coupling

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Abstract

J1610+1811 is a blazer-class quasar at $z = 6.5$ (lookback time ~ 12.9 Gyr) with a relativistic jet Lorentz factor $\Gamma \approx 4.5$. UQFF introduces a relativistic coupling constant $k_{\text{rel}} = \Gamma^2 = k_{\text{rel},0} \times 20.25$ (Lorentz factor squared) to scale the vacuum buoyancy force in the jet frame. The Friedmann Hubble parameter $H(z = 6.5)$ is computed from $H(z) = H_0\sqrt{0.3(1+z)^3 + 0.7}$, and $F_{U_Bi_i} \approx -8.32 \times 10^{217}$ N is evaluated in the observer frame.

2. Core Physics

2.1 Relativistic UQFF Coupling

$$k_{\text{rel}} = \Gamma^2 = k_{\text{rel},0} \times \Gamma^2 = k_{\text{rel},0} \times (4.5)^2 = 20.25 \cdot k_{\text{rel},0}$$

The Lorentz factor squared enhancement represents the relativistic Doppler amplification of the vacuum buoyancy force in the jet frame.

2.2 Friedmann $H(z)$ at $z = 6.5$

$$H(z) = H_0\sqrt{0.3(1+z)^3 + 0.7}$$

For $z = 6.5$:

$$(1 + 6.5)^3 = 7.5^3 = 421.875$$

$$H(6.5) = H_0\sqrt{0.3 \times 421.875 + 0.7} = H_0\sqrt{126.56 + 0.7} = H_0\sqrt{127.26}$$

$$H(6.5) \approx 11.3H_0 = 11.3 \times 67.4 \text{ km/s/Mpc} \approx 761 \text{ km/s/Mpc}$$

2.3 Relativistic $F_{U_Bi_i}$

$$F_{U_Bi_i}^{\text{relativistic}} = F_{U_Bi_i}^{\text{standard}} \cdot k_{\text{rel}} = -8.32 \times 10^{217} \times 20.25 \approx -1.69 \times 10^{219} \text{ N}$$

(in the jet co-moving frame; observer-frame is un-boosted)

2.4 Lookback Time at $z = 6.5$

$$t_{\text{lookback}} \approx 12.9 \text{ Gyr}$$

The Universe was only ~ 820 Myr old when this jet was active — the UQFF vacuum energy density was $\rho_{\text{vac}}(z=6.5) = \rho_{\text{vac},0} \cdot (1+z)^\alpha$, higher than today by the UQFF expansion index α .

3. Key Values

Quantity	Formula	Value
z	Photometric/spectro	6.5
Γ	VLBI jet model	4.5
k_{rel}	Γ^2	20.25
$H(z=6.5)$	Friedmann	$\sim 761 \text{ km/s/Mpc}$
$F_{\text{U_Bi_i}}$ (obs frame)	Standard	$-8.32 \times 10^{217} \text{ N}$
$F_{\text{U_Bi_i}}$ (jet frame)	$\times k_{\text{rel}}$	$-1.69 \times 10^{219} \text{ N}$

4. Physical Significance

J1610+1811 at $z = 6.5$ presents UQFF at the earliest cosmic epoch in the dataset. The $k_{\text{rel}} = \Gamma^2$ coupling is the first relativistic enhancement factor in UQFF AGN physics — it predicts that high- Γ relativistic jets experience systematically larger UQFF vacuum buoyancy forces in their rest frame. This has cosmological implications: early universe ($z > 5$) AGN jets would have experienced larger vacuum buoyancy during the epoch of reionization, potentially accelerating the growth of early massive black holes — addressing the “first quasar” problem.

5. Deduplication Note

- **vs. PAPER_346-350 (low- z AGN):** All earlier AGN papers in this series have $z < 1.5$; J1610 at $z = 6.5$ is $6.5\times$ higher redshift.
 - **vs. PAPER_360 vs. k_{rel} :** No earlier UQFF paper includes the Lorentz factor Γ^2 relativistic boost.
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6. Classification

Physics Territory: FIRST UQFF high- z quasar jet ($z=6.5$) with Γ^2 relativistic coupling and Friedmann $H(z)$

Scale: Cosmological ($z = 6.5$, lookback ~ 12.9 Gyr)

CP Implementation: J1610HighZQuasarJetFUBiCalculator (CondensedPhysics4.py,
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